AERIS Expert Review Packet

Anomaly

Source / metric	openaq / Ozone (ozone)
Observed / expected baseline	0.069 / 0.0169615 ppm
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-08-02 18:00 UTC
Location	29.6256, -95.2672
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	8 / 779	20.95 to 100; mean 68.8698	34.21 at 2026-08-02 17:55Z; 5 min before event
asos	Air temperature (temperature)	degC	8 / 779	23.8889 to 37; mean 30.3363	34 at 2026-08-02 17:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	8 / 1026	0 to 360; mean 180.984	320 at 2026-08-02 17:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	8 / 1061	0 to 33.4389; mean 2.9242	2.57222 at 2026-08-02 17:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	11 / 143	5875.21 to 5946.43; mean 5902.1	5885.49 at 2026-08-02 18:00Z; at event time
noaa_gfs	Planetary boundary layer height (pbl_height)	m	11 / 143	11.2803 to 2539.72; mean 825.819	2354.3 at 2026-08-02 18:00Z; at event time
noaa_gfs	Precipitable water (precipitable_water)	mm	11 / 143	26.8117 to 57.7464; mean 39.5761	32.3202 at 2026-08-02 18:00Z; at event time
noaa_gfs	Surface pressure (surface_pressure)	hPa	11 / 143	1003.4 to 1013.29; mean 1008.61	1009.19 at 2026-08-02 18:00Z; at event time
noaa_gfs	850-hPa temperature (t_850)	degC	11 / 143	18.2063 to 22.5646; mean 20.2624	19.5048 at 2026-08-02 18:00Z; at event time
noaa_gfs	10-m eastward wind (u_10m)	m/s	11 / 143	-3.62395 to 4.5116; mean 0.9388	-1.43038 at 2026-08-02 18:00Z; at event time
noaa_gfs	10-m northward wind (v_10m)	m/s	11 / 143	-3.70918 to 6.32564; mean 1.1229	-2.72918 at 2026-08-02 18:00Z; at event time
openaq	Ozone (ozone)	ppm	17 / 1108	0 to 0.081; mean 0.0291	0.069 at 2026-08-02 18:00Z; at event time
openaq	PM10 (pm10)	ug/m3	3 / 217	11 to 153; mean 28.4009	17 at 2026-08-02 18:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM2.5 (pm25)	ug/m3	71 / 2786	0 to 50.8; mean 8.1906	5.2 at 2026-08-02 18:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 360	0 to 100; mean 27.025	96 at 2026-08-02 18:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 360	34 to 96; mean 70.0806	47 at 2026-08-02 18:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 360	0 to 0.22; mean 0.0009	0 at 2026-08-02 18:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 360	1007 to 1014; mean 1010.8	1011 at 2026-08-02 18:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 360	25.3 to 37.65; mean 31.1703	35.32 at 2026-08-02 18:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 360	0 to 355; mean 182.858	343 at 2026-08-02 18:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 360	0.26 to 5.85; mean 2.1516	3.11 at 2026-08-02 18:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	71 / 5006	0 to 313; mean 10.5687	12.5 at 2026-08-02 18:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	CO column (s5p_co_column)	mol/m^2	8 / 8	0.0275078 to 0.0328455; mean 0.0306	0.0328455 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	5 / 5	0.000197921 to 0.000318768; mean 0.0003	0.000306598 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	4 / 4	4.01789e-05 to 5.22023e-05; mean 0	4.15036e-05 at 2026-08-02 19:10Z; 1 h 10 min after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-08-02 19:10Z; 1 h 10 min after event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	5 / 5	-6.07222e-05 to 0.000178122; mean 0	-6.07222e-05 at 2026-08-02 18:38Z; 39 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-08-02 18:38Z; 39 min after event
tceq	Carbon monoxide (co)	ppm	3 / 188	0 to 0.7; mean 0.1957	0.2 at 2026-08-02 17:00Z; 1 h before event
tceq	Nitrogen dioxide (no2)	ppb	12 / 735	-2.4 to 33.6; mean 4.5578	4.2 at 2026-08-02 18:00Z; at event time
tceq	Sulfur dioxide (so2)	ppb	3 / 187	-0.6 to 7.2; mean -0.0214	-0.2 at 2026-08-02 18:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	08-01 06Z 92.17, 12Z 74.32, 18Z 49.89, 08-02 00Z 75.74, 06Z 87.39, 12Z 61.93, 18Z 36.74, 08-03 00Z 70.06, 06Z 88.9, 12Z 68.67, 18Z 46.4, 08-04 00Z 75.82
asos	Air temperature (temperature)	08-01 06Z 25.84, 12Z 30.26, 18Z 34.18, 08-02 00Z 30.04, 06Z 27.87, 12Z 31.06, 18Z 35.13, 08-03 00Z 29.49, 06Z 26.25, 12Z 30.64, 18Z 34.42, 08-04 00Z 28.5
asos	Wind direction (wind_direction)	08-01 06Z 170.9, 12Z 235.2, 18Z 233.5, 08-02 00Z 211.6, 06Z 237.4, 12Z 127.4, 18Z 179.2, 08-03 00Z 164.5, 06Z 140.4, 12Z 158.8, 18Z 155.6, 08-04 00Z 160.5
asos	Wind speed (wind_speed)	08-01 06Z 2.194, 12Z 4.162, 18Z 3.35, 08-02 00Z 3.058, 06Z 2.814, 12Z 1.961, 18Z 2.919, 08-03 00Z 2.942, 06Z 1.526, 12Z 2.166, 18Z 4.485, 08-04 00Z 3.486
noaa_gfs	500-hPa geopotential height (gh_500)	08-01 06Z 5945, 12Z 5927, 18Z 5927, 08-02 00Z 5904, 06Z 5901, 12Z 5881, 18Z 5886, 08-03 00Z 5877, 06Z 5887, 12Z 5881, 18Z 5896, 08-04 00Z 5900, 06Z 5914
noaa_gfs	Planetary boundary layer height (pbl_height)	08-01 06Z 409.8, 12Z 368.6, 18Z 1621, 08-02 00Z 881.9, 06Z 310.5, 12Z 193.7, 18Z 2125, 08-03 00Z 1028, 06Z 170.7, 12Z 94.33, 18Z 2232, 08-04 00Z 992.9, 06Z 306.3
noaa_gfs	Precipitable water (precipitable_water)	08-01 06Z 38.77, 12Z 42.33, 18Z 50.08, 08-02 00Z 56.54, 06Z 52.17, 12Z 40.6, 18Z 32.29, 08-03 00Z 29.79, 06Z 31.64, 12Z 33.03, 18Z 34.55, 08-04 00Z 39.07, 06Z 33.63
noaa_gfs	Surface pressure (surface_pressure)	08-01 06Z 1011, 12Z 1011, 18Z 1010, 08-02 00Z 1006, 06Z 1008, 12Z 1008, 18Z 1008, 08-03 00Z 1006, 06Z 1009, 12Z 1009, 18Z 1009, 08-04 00Z 1008, 06Z 1010
noaa_gfs	850-hPa temperature (t_850)	08-01 06Z 22.18, 12Z 21.94, 18Z 19.69, 08-02 00Z 21.56, 06Z 20.69, 12Z 20.46, 18Z 19.41, 08-03 00Z 20.75, 06Z 19.34, 12Z 18.52, 18Z 18.91, 08-04 00Z 19.94, 06Z 20.01
noaa_gfs	10-m eastward wind (u_10m)	08-01 06Z 1.591, 12Z 2.248, 18Z 4.068, 08-02 00Z 1.758, 06Z 2.977, 12Z 1.254, 18Z -0.3686, 08-03 00Z -1.179, 06Z 0.9857, 12Z 1.19, 18Z -0.7632, 08-04 00Z -2.216, 06Z 0.659
noaa_gfs	10-m northward wind (v_10m)	08-01 06Z 3.249, 12Z 1.848, 18Z 0.5632, 08-02 00Z 0.3482, 06Z 1.705, 12Z -1.725, 18Z -2.846, 08-03 00Z -0.9135, 06Z 2.355, 12Z 0.8952, 18Z 1.676, 08-04 00Z 4.509, 06Z 2.934
openaq	Ozone (ozone)	08-01 06Z 0.01185, 12Z 0.01425, 18Z 0.03155, 08-02 00Z 0.0223, 06Z 0.008851, 12Z 0.024, 18Z 0.05939, 08-03 00Z 0.05227, 06Z 0.02463, 12Z 0.02048, 18Z 0.05143, 08-04 00Z 0.02236, 06Z 0.01433
openaq	PM10 (pm10)	08-01 06Z 16.94, 12Z 30.22, 18Z 31.83, 08-02 00Z 24.33, 06Z 22.22, 12Z 22.89, 18Z 16.61, 08-03 00Z 20.44, 06Z 20.5, 12Z 50.18, 18Z 43.35, 08-04 00Z 41.67, 06Z 38.33
openaq	PM2.5 (pm25)	08-01 06Z 3.341, 12Z 4.897, 18Z 5.655, 08-02 00Z 8.785, 06Z 9.929, 12Z 8.068, 18Z 8.749, 08-03 00Z 10.32, 06Z 9.128, 12Z 9.872, 18Z 12.05, 08-04 00Z 11.63, 06Z 14.92
openweather	Cloud cover (cloud_cover)	08-01 06Z 4.8, 12Z 13.1, 18Z 18.45, 08-02 00Z 55.87, 06Z 67.31, 12Z 85.45, 18Z 32.53, 08-03 00Z 19.73, 06Z 1.069, 12Z 0, 18Z 2.1, 08-04 00Z 23.13
openweather	Relative humidity (humidity)	08-01 06Z 87.5, 12Z 81.24, 18Z 55.77, 08-02 00Z 71.87, 06Z 85.97, 12Z 70.94, 18Z 41.77, 08-03 00Z 65.2, 06Z 83.55, 12Z 75.97, 18Z 51.1, 08-04 00Z 71.7
openweather	Precipitation rate (precipitation)	08-01 06Z 0, 12Z 0, 18Z 0.003871, 08-02 00Z 0, 06Z 0.007586, 12Z 0, 18Z 0, 08-03 00Z 0, 06Z 0, 12Z 0, 18Z 0, 08-04 00Z 0
openweather	Surface pressure (pressure)	08-01 06Z 1013, 12Z 1014, 18Z 1010, 08-02 00Z 1009, 06Z 1010, 12Z 1012, 18Z 1009, 08-03 00Z 1009, 06Z 1011, 12Z 1012, 18Z 1010, 08-04 00Z 1011
openweather	Air temperature (temperature)	08-01 06Z 26.73, 12Z 29.56, 18Z 35.64, 08-02 00Z 31.39, 06Z 28.4, 12Z 30.66, 18Z 36.36, 08-03 00Z 31.37, 06Z 27.22, 12Z 30.23, 18Z 35.93, 08-04 00Z 30.17
openweather	Wind direction (wind_direction)	08-01 06Z 154.6, 12Z 232.2, 18Z 264, 08-02 00Z 230.7, 06Z 194.6, 12Z 90.61, 18Z 166.6, 08-03 00Z 158.4, 06Z 175.3, 12Z 215.5, 18Z 151.8, 08-04 00Z 161
openweather	Wind speed (wind_speed)	08-01 06Z 2.446, 12Z 2.695, 18Z 2.235, 08-02 00Z 2.286, 06Z 2.043, 12Z 1.942, 18Z 2.036, 08-03 00Z 1.967, 06Z 1.916, 12Z 1.322, 18Z 2.397, 08-04 00Z 2.572
purpleair	PM2.5 (pm25)	08-01 06Z 4.426, 12Z 6.305, 18Z 7.207, 08-02 00Z 12.67, 06Z 12.22, 12Z 10.35, 18Z 9.443, 08-03 00Z 12.42, 06Z 12.13, 12Z 12.35, 18Z 13.93, 08-04 00Z 13.16
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1

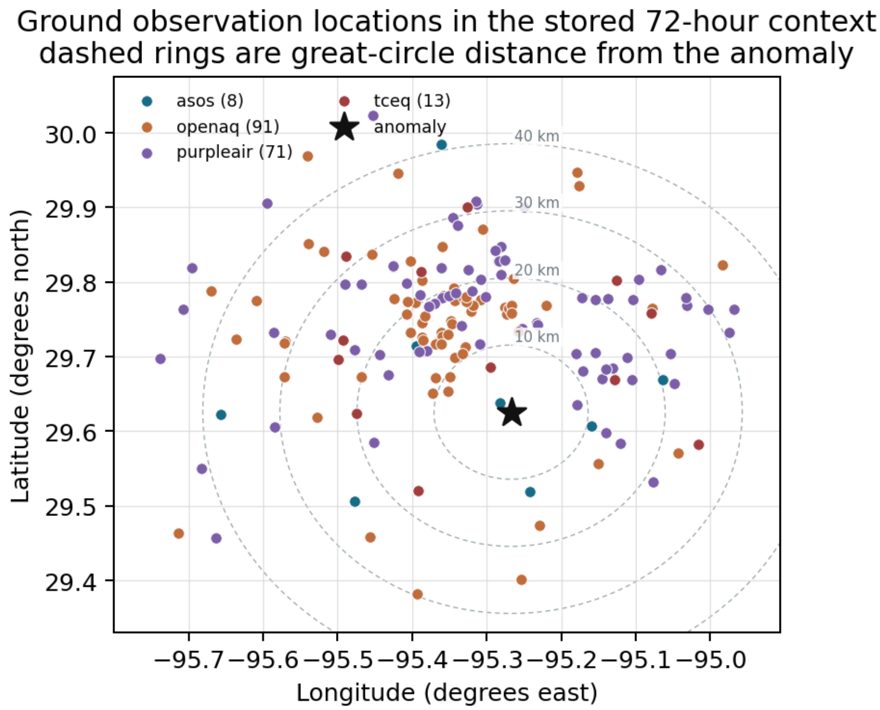
Source	Metric	Values
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	CO column (s5p_co_column)	08-01 18Z 0.02751, 08-02 18Z 0.03284, 08-03 18Z 0.03095
sentinel5p	CO granules available (s5p_co_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	08-01 18Z 0.0001979, 08-02 18Z 0.0003127, 08-03 18Z 0.0002755
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	08-01 18Z 4.018e-05, 08-02 18Z 4.15e-05, 08-03 18Z 4.973e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	08-01 18Z 6.519e-06, 08-02 18Z -4.61e-05, 08-03 18Z 0.0001056
sentinel5p	SO2 granules available (s5p_so2_granule_available)	08-01 18Z 1, 08-02 18Z 1, 08-03 18Z 1
tceq	Carbon monoxide (co)	08-01 06Z 0.1611, 12Z 0.2056, 18Z 0.1812, 08-02 00Z 0.22, 06Z 0.1167, 12Z 0.1333, 18Z 0.16, 08-03 00Z 0.2444, 06Z 0.2389, 12Z 0.2722, 18Z 0.25, 08-04 00Z 0.2111, 06Z 0.1333
tceq	Nitrogen dioxide (no2)	08-01 06Z 3.42, 12Z 3.552, 18Z 4.337, 08-02 00Z 6.082, 06Z 3.036, 12Z 3.271, 18Z 2.811, 08-03 00Z 4.778, 06Z 8.049, 12Z 7.837, 18Z 4.965, 08-04 00Z 2.549, 06Z 2.683
tceq	Sulfur dioxide (so2)	08-01 06Z -0.1944, 12Z -0.1167, 18Z -0.04706, 08-02 00Z -0.18, 06Z -0.07778, 12Z -0.09444, 18Z -0.02778, 08-03 00Z -0.03333, 06Z -0.2056, 12Z 0.7444, 18Z -0.07222, 08-04 00Z -0.04444, 06Z -0.2667

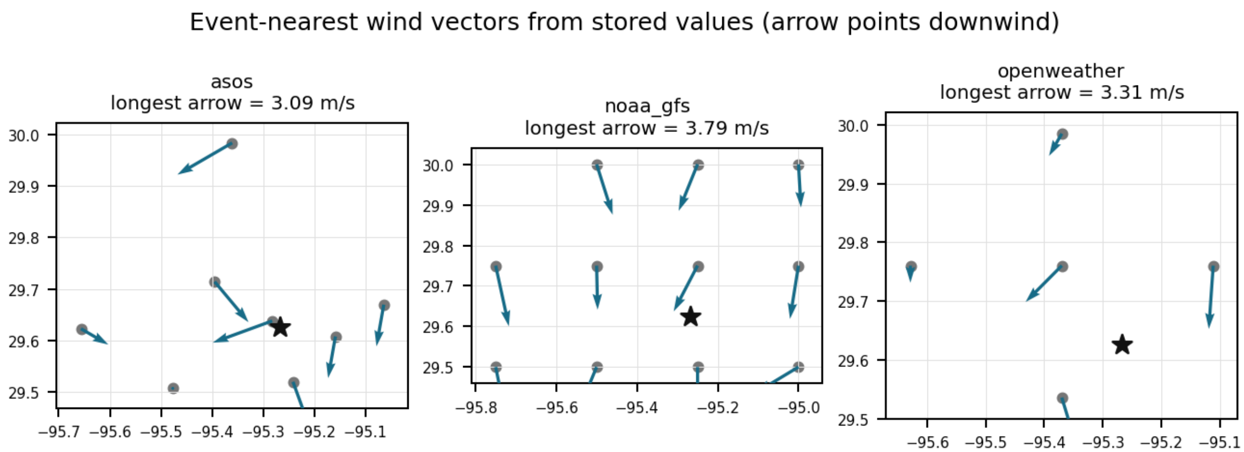
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	Relative humidity (humidity)	HOU (1.977 km) 69.1, EFD (10.684 km) 67.37, LVJ (12.118 km) 71.43, MCJ (15.782 km) 66.34, T41 (20.217 km) 70.28, AXH (24.247 km) 68.99, SGR (37.632 km) 73.47, IAH (40.904 km) 64.7
asos	Air temperature (temperature)	HOU (1.977 km) 30.54, EFD (10.684 km) 30.84, LVJ (12.118 km) 30.14, MCJ (15.782 km) 30.58, T41 (20.217 km) 30.17, AXH (24.247 km) 29.95, SGR (37.632 km) 29.88, IAH (40.904 km) 30.99
asos	Wind direction (wind_direction)	HOU (1.977 km) 205.2, EFD (10.684 km) 174.5, LVJ (12.118 km) 163.9, MCJ (15.782 km) 204.4, T41 (20.217 km) 196.8, AXH (24.247 km) 107.2, SGR (37.632 km) 207, IAH (40.904 km) 190.3
asos	Wind speed (wind_speed)	HOU (1.977 km) 3.569, EFD (10.684 km) 4.29, LVJ (12.118 km) 2.19, MCJ (15.782 km) 3.651, T41 (20.217 km) 3.454, AXH (24.247 km) 1.133, SGR (37.632 km) 2.897, IAH (40.904 km) 3.04
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (13.932 km) 5902, gfs:29.50,-95.25 (14.065 km) 5902, gfs:29.75,-95.50 (26.402 km) 5902, gfs:29.50,-95.50 (26.496 km) 5903, gfs:29.75,-95.00 (29.284 km) 5901, gfs:29.50,-95.00 (29.376 km) 5901, gfs:30.00,-95.25 (41.664 km) 5902, gfs:30.00,-95.50 (47.304 km) 5902, +3 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (13.932 km) 1009, gfs:29.50,-95.25 (14.065 km) 703.5, gfs:29.75,-95.50 (26.402 km) 1085, gfs:29.50,-95.50 (26.496 km) 760, gfs:29.75,-95.00 (29.284 km) 796.7, gfs:29.50,-95.00 (29.376 km) 709.9, gfs:30.00,-95.25 (41.664 km) 958.6, gfs:30.00,-95.50 (47.304 km) 1117, +3 more
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (13.932 km) 39.98, gfs:29.50,-95.25 (14.065 km) 40.62, gfs:29.75,-95.50 (26.402 km) 39.54, gfs:29.50,-95.50 (26.496 km) 40.18, gfs:29.75,-95.00 (29.284 km) 39.83, gfs:29.50,-95.00 (29.376 km) 40.47, gfs:30.00,-95.25 (41.664 km) 38.82, gfs:30.00,-95.50 (47.304 km) 38.29, +3 more

Source	Metric	Values
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (13.932 km) 1009, gfs:29.50,-95.25 (14.065 km) 1010, gfs:29.75,-95.50 (26.402 km) 1008, gfs:29.50,-95.50 (26.496 km) 1009, gfs:29.75,-95.00 (29.284 km) 1010, gfs:29.50,-95.00 (29.376 km) 1011, gfs:30.00,-95.25 (41.664 km) 1008, gfs:30.00,-95.50 (47.304 km) 1006, +3 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (13.932 km) 20.16, gfs:29.50,-95.25 (14.065 km) 20.21, gfs:29.75,-95.50 (26.402 km) 20.3, gfs:29.50,-95.50 (26.496 km) 20.34, gfs:29.75,-95.00 (29.284 km) 20.09, gfs:29.50,-95.00 (29.376 km) 20.13, gfs:30.00,-95.25 (41.664 km) 20.25, gfs:30.00,-95.50 (47.304 km) 20.49, +3 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (13.932 km) 0.7845, gfs:29.50,-95.25 (14.065 km) 0.9283, gfs:29.75,-95.50 (26.402 km) 1.098, gfs:29.50,-95.50 (26.496 km) 0.9337, gfs:29.75,-95.00 (29.284 km) 0.7514, gfs:29.50,-95.00 (29.376 km) 1.083, gfs:30.00,-95.25 (41.664 km) 0.3429, gfs:30.00,-95.50 (47.304 km) 1.032, +3 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (13.932 km) 1.06, gfs:29.50,-95.25 (14.065 km) 1.656, gfs:29.75,-95.50 (26.402 km) 1.067, gfs:29.50,-95.50 (26.496 km) 1.345, gfs:29.75,-95.00 (29.284 km) 0.815, gfs:29.50,-95.00 (29.376 km) 2.181, gfs:30.00,-95.25 (41.664 km) 0.688, gfs:30.00,-95.50 (47.304 km) 0.8473, +3 more
openaq	Ozone (ozone)	320 (0.0 km) 0.0269, 325 (7.264 km) 0.0125, 3378 (12.079 km) 0.02743, 332 (14.284 km) 0.03184, 2039 (16.472 km) 0.02796, 3055 (16.85 km) 0.02859, 312 (20.009 km) 0.0303, 1319333 (20.557 km) 0.03497, +9 more
openaq	PM10 (pm10)	13380082 (12.079 km) 41.51, 271 (14.284 km) 25.13, 15852419 (16.681 km) 18.48
openaq	PM2.5 (pm25)	14157975 (0.003 km) 8.496, 14140751 (7.245 km) 7.951, 15472101 (8.787 km) 5.806, 13418566 (9.543 km) 7.463, 8967021 (10.575 km) 8.895, 17070153 (10.704 km) 8.568, 16564662 (10.96 km) 8.71, 9416627 (11.119 km) 5.518, +63 more
openweather	Cloud cover (cloud_cover)	grid:south (14.076 km) 31.94, grid:center (17.969 km) 26.07, grid:east (21.261 km) 27.28, grid:west (37.982 km) 28.04, grid:north (41.171 km) 21.79
openweather	Relative humidity (humidity)	grid:south (14.076 km) 70.89, grid:center (17.969 km) 68.6, grid:east (21.261 km) 74.01, grid:west (37.982 km) 69.46, grid:north (41.171 km) 67.44
openweather	Precipitation rate (precipitation)	grid:south (14.076 km) 0, grid:center (17.969 km) 0, grid:east (21.261 km) 0.003056, grid:west (37.982 km) 0.001667, grid:north (41.171 km) 0
openweather	Surface pressure (pressure)	grid:south (14.076 km) 1011, grid:center (17.969 km) 1011, grid:east (21.261 km) 1011, grid:west (37.982 km) 1011, grid:north (41.171 km) 1011
openweather	Air temperature (temperature)	grid:south (14.076 km) 30.95, grid:center (17.969 km) 31.65, grid:east (21.261 km) 30.86, grid:west (37.982 km) 31.14, grid:north (41.171 km) 31.26
openweather	Wind direction (wind_direction)	grid:south (14.076 km) 213.3, grid:center (17.969 km) 200, grid:east (21.261 km) 172.3, grid:west (37.982 km) 189.7, grid:north (41.171 km) 139
openweather	Wind speed (wind_speed)	grid:south (14.076 km) 2.159, grid:center (17.969 km) 2.077, grid:east (21.261 km) 3.103, grid:west (37.982 km) 1.383, grid:north (41.171 km) 2.037
purpleair	PM2.5 (pm25)	182191 (8.594 km) 12.02, 6752 (10.934 km) 7.511, 186263 (11.178 km) 12.14, 235425 (12.183 km) 10.79, 99797 (12.499 km) 10.25, 247283 (12.722 km) 11.53, 128007 (12.806 km) 4.508, 97279 (13.522 km) 12.66, +63 more
tceq	Carbon monoxide (co)	482011035 (12.059 km) 0.1667, 482011039 (14.272 km) 0.1475, 482011052 (24.007 km) 0.2703
tceq	Nitrogen dioxide (no2)	482010416 (7.255 km) 4.703, 482011035 (12.059 km) 7.105, 482011039 (14.272 km) 3.05, 480391004 (16.839 km) 1.683, 482011015 (23.434 km) 7.448, 482010055 (23.737 km) 2.61, 482010026 (23.979 km) 5.805, 482011052 (24.007 km) 9.068, +4 more
tceq	Sulfur dioxide (so2)	482011035 (12.059 km) -0.2825, 482011039 (14.272 km) 0.4836, 482010051 (20.024 km) -0.2492


Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly
asos (8)
openaq (91)
purpleair (71)
tceq (13)
anomaly
10 km
20 km
30 km
40 km
Latitude (degrees north)
Longitude (degrees east)
30.0
29.9
29.8
29.7
29.6
29.5
29.4
−95.7
−95.6
−95.5
−95.4
−95.3
−95.2
−95.1
−95.0


Event-nearest wind vectors from stored values (arrow points downwind)
asos
longest arrow = 3.09 m/s
noaa_gfs
longest arrow = 3.79 m/s
openweather
longest arrow = 3.31 m/s

Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

The pollutant is particulate matter (PM2.5) with a concentration of 12.5 ug/m3, which is elevated compared to the mean value of 10.5687 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 36

Strong photochemical ozone production was a likely primary cause because the anomaly occurred during very hot afternoon conditions with unusually low humidity and a sharp daytime ozone rise, which favors rapid ozone formation from Houston-area NOx and VOC precursors.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 36

Photochemical ozone production was strongly favored near Houston at 2026-08-02 18:00 UTC because surface temperatures were very high at about 35 to 36 C while relative humidity fell to about 34 to 47 percent, conditions consistent with strong solar-driven ozone formation after morning moist conditions gave way to a much hotter and drier afternoon.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 36

Satellite measurements from sentinel5p showing elevated formaldehyde column densities along with ground-level nitrogen dioxide measurements from tceq demonstrate the availability of reactive precursor gases required for high ozone formation rates.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 36

The PM2.5 levels were significantly higher than usual, with readings up to 12.02 ug/m3 at sensor 182191.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 36

Smoke, dust, and sulfur-plume forcing is not the leading explanation for the Houston ozone anomaly because PM2.5 remained modest near about 5 to 13 ug/m3, PM10 near the event was 17 ug/m3, sulfur dioxide was near zero or negative, and satellite SO2 showed no positive plume signal.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 36

Atmospheric conditions during the ozone anomaly were characterized by high surface temperatures near 35°C to 36°C, low relative humidity around 36%, and a planetary boundary layer height above 2,100 meters.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 36

The air quality in Houston, Texas is poor due to high levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 36

Regional ground-level ozone across 17 monitoring sensors in the Houston area peaked during the 6-hour period around 18:00 UTC on August 2, 2026, reaching a network average of 0.059 ppm.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 36

High ambient temperatures exceeding 35°C combined with low relative humidity created optimal environmental conditions for rapid photochemical production of surface ozone in Houston on August 2, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 36

A smoke, dust, or sulfur-rich plume was not the leading driver of the ozone anomaly because particulate matter remained modest near the event, sulfur dioxide was near zero, and satellite sulfur dioxide was not elevated, although satellite carbon monoxide and formaldehyde were somewhat elevated and therefore remain compatible with urban-industrial or combustion-related ozone precursors rather than a primary aerosol event.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 36

NO2 levels were also elevated, with readings up to 4.2 ppb at monitor 482010416.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 36

Weak-to-moderate and shifting low-level winds were a likely contributing cause because variable flow near the anomaly time can recirculate and concentrate urban and petrochemical precursor emissions over east-central Houston instead of ventilating them efficiently.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 36

The air-quality anomaly in Houston, Texas on August 2nd was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 36

The anomaly occurred at 2026-08-02T18:00:00+00:00 (0.0 min away) and is located approximately 8.594 km from the nearest sensor.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 36

A deep planetary boundary layer exceeding 2,000 meters in height under light surface winds facilitated the vertical mixing and accumulation of ozone and its precursors during peak daytime heating.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 36

The openaq ozone measurement at 2026-08-02T18:00:00+00:00 was 0.069 ppm, compared with an expected value of 0.016961538461538462 ppm and a z-score of 6.033228617906318, which was classified as severe.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 36

Elevated surface temperatures exceeding 35 degrees Celsius combined with relative humidity dropping below 37 percent at 18:00 UTC created optimal atmospheric conditions for rapid photochemical ozone formation measured by openaq.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 36

Area-wide openaq ozone was elevated during the 2026-08-02 18Z 6-hour period, with a mean of 0.05939 ppm across 17 sensors, higher than the context-window mean of 0.0291 ppm.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 36

Regional stagnation and boundary-layer conditions were a likely contributing cause because a deep afternoon planetary boundary layer under summertime high-pressure conditions can mix residual and upwind precursor-rich air into the surface layer while maintaining a broad elevated ozone episode, and the low particulate and sulfur signals argue against smoke, dust, or SO2-rich plumes as the dominant explanation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 36

The temperature at the time of the anomaly was 35.32 degC, which is higher than the mean value of 31.1703 degC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 36

The temperature is unusually high, contributing to the air-quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 36

An ozone monitoring sensor at coordinates (29.626, -95.267) in Houston recorded an anomalous surface ozone concentration of 0.069 ppm at 2026-08-02T18:00:00+00:00, compared to an expected baseline of approximately 0.017 ppm.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 36

Satellite retrievals over Houston on August 2, 2026, demonstrated elevated atmospheric column amounts of formaldehyde and carbon monoxide relative to preceding days.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 36

Meteorological dispersion and transport conditions could have concentrated locally produced ozone and its precursors near east-central Houston because winds were light at roughly 2 to 3 m/s, wind direction shifted markedly from easterly or northerly components into the afternoon, and modeled planetary boundary layer height increased to about 2350 m, a pattern consistent with recirculation or mixing down of polluted air rather than rapid ventilation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 36

Houston meteorology at 18:00 UTC on 2026-08-02 favored mixing and recirculation of ozone precursors because the boundary-layer height was about 2354 m while surface winds were light at about 2 to 3 m/s and wind direction shifted around the period.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 36

Surface ozone concentrations in Houston reached an anomalous peak of 0.069 ppm at 18:00 UTC on August 2, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 36

The wind direction and speed are not typical for this time of year, potentially dispersing pollutants in a way that exacerbates the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 36

The observed pollutant anomaly was ozone at coordinates 29.626, -95.267 in Houston at 2026-08-02T18:00:00+00:00.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 36

The air quality in Houston, Texas is poor due to high levels of SO_2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 36

The air quality in Houston, Texas is poor due to high levels of NO2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 36

There is an unusual concentration of particulate matter (PM2.5) in the air, likely caused by industrial or vehicular emissions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 36

Houston ozone at 18:00 UTC on 2026-08-02 is most consistent with enhanced photochemical production because area ozone rose to a 6-hour mean of 0.05939 ppm while afternoon temperature reached about 35 to 36 C and humidity fell to about 37 to 42 percent near the event.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 36

Planetary boundary layer heights exceeding 2100 meters modeled by gfs at 18:00 UTC facilitated deep atmospheric boundary layer mixing during peak solar radiation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 36

Elevated regional formaldehyde and nitrogen dioxide precursor columns provided necessary chemical feedstocks for photochemical ozone formation near the anomaly location.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 36 of 36

Surface meteorological observations at 18:00 UTC on August 2, 2026, recorded high temperatures between 35.1 degC and 36.4 degC accompanied by low relative humidity around 34 percent to 37 percent near the anomaly location.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.